2-26-26 (WEEK 1)

Shun/翔海 (@shun4midx)

PROPOSITION (PASCAL'S IDENTITY)

$\forall\, 1 \le r \le n$, $\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$

Proof: ① $\binom{n}{r} + \binom{n}{r-1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!} = \frac{n!}{(r-1)!(n-r)!}\left[\frac{1}{r} + \frac{1}{n-r+1}\right] = \frac{(n+1)!}{r!(n-r+1)!} = \binom{n+1}{r}$ □

② Recall, $\binom{n}{r}$ = # subsets of size r from a set of size n, say $[n]$.
Let $S = \binom{[n]}{r}$ be the set of these subsets, of course $|S| = \binom{n}{r}$

Let $S_1 = \{Y \in \binom{[n]}{r} \mid n \notin Y\}$ and $S_2 = \{Y \in \binom{[n]}{r} \mid n \in Y\}$, then $S = S_1 \sqcup S_2$
Observe, $S_1 = \binom{[n-1]}{r} \Rightarrow |S_1| = \binom{n-1}{r}$
$S_2 = \{Y \subseteq [n] \mid n \in Y,\ |Y| = r\} = \{Y \cup \{n\} \mid Y \subseteq [n-1],\ |Y| = r-1\} \Rightarrow |S_2| = \binom{n-1}{r-1}$ since S_2 is a bijection of $\binom{[n-1]}{r}$

$\therefore$ By the sum rule, $|S| = |S_1| + |S_2| \Rightarrow \binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ □

TAKEAWAY

$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ gives a recursive formula for computing $\binom{n}{r}$, which is simpler than $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

PASCAL'S TRIANGLE

$$
\begin{array}{ccccccc}
 & & & 1\ \binom{0}{0} & & & \\
 & & 1\ \binom{1}{0} & & \binom{1}{1}\ 1 & & \\
 & 1\ \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2}\ 1 & \\
1\ \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3}\ 1 \\
 & & & \vdots & & &
\end{array}
$$

THEOREM (BINOMIAL THEOREM)

For any $n \in \mathbb{N}_0$, $x, y \in \mathbb{F}$, $(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}$

Proof

Notice, $(x+y)^n = \underbrace{(x+y)\cdots(x+y)}_{n}$

Each monomial comes from choosing one term from each factor $(x+y)$.
They must take the form $x^r y^{n-r}$, where r is the # factors which come from choosing x, $0 \le r \le n$

$\therefore$ Coef of $x^r y^{n-r}$ = # ways of choosing x r times

By def, there are $\binom{n}{r}$ ways to choose r factors to choose x, choose y for the rest
$\therefore$ Coef of $x^r y^{n-r} = \binom{n}{r}$, i.e. $(x+y)^n = \sum_{r=0}^{n} \binom{n}{r} x^r y^{n-r}$ □

COROLLARY

The total # subsets of an n element set $= 2^n$

Proof

① Apply the product rule for each of the n elements, ask if it is in the subset or not.
$\because \exists$ 2 options per question
$\therefore 2^n$ subsets □

② Let S be the set of all subsets.
Then, for $0 \le r \le n$, let $S_r \subseteq S$ be the subset of S with elements size r
$\therefore S = S_0 \cup \ldots \sqcup S_r \Rightarrow$ By the sum rule, $|S| = \sum_{r=0}^{n} |S_r| = \sum_{r=0}^{n} \binom{n}{r}$
$\therefore |S| = \sum_{r=0}^{n} \binom{n}{r} 1^r 1^{n-r} = (1+1)^n = 2^n$ □
↑ bin thm

Shun/翔海 (@shun4midx)

COROLLARY

Let X be an n element set, $n \geq 1$.

Then, #even-sized subsets = #odd-sized subsets (of X)

Proof

Want to show: $\sum_{r:\text{even}} \binom{n}{r} - \sum_{r:\text{odd}} \binom{n}{r} = 0$

Notice, $\sum_{r:\text{even}} \binom{n}{r} - \sum_{r:\text{odd}} \binom{n}{r} = \sum_{r=0}^{n} \binom{n}{r}(-1)^r = \sum_{r=0}^{n} \binom{n}{r}(-1)^r 1^{n-r} \overset{\text{bin thm}}{=} (-1+1)^r = 0$ □

EXAMPLE

In poker, we are dealt a hand of five cards from a standard deck.

What is the probability of a full-house? (3-of-a-kind + a pair, e.g. 33QQQ)

Solution (w/o order (∵ cards are chosen w/o replacement))

$S = \{$subsets of 5 cards from the deck$\}$

Say D = deck of cards, then $S = \binom{D}{5} \Rightarrow |S| = \binom{52}{5}$

$E = \{$full houses$\} = \{$triples + pairs$\}$

$\therefore P(\text{full house}) = \frac{|E|}{|S|}$

Consider $|E|$,

① Choose triple: Choose value (13 options), choose suits $\binom{4}{3}$ options

② Choose pair: Choose value (12 options), choose suits $\binom{4}{2}$ options

By the product rule, # triples $= 13\binom{4}{3}$, # pairs $= 12\binom{4}{2} \Rightarrow$ # full houses $= 13\binom{4}{3}12\binom{4}{2} = |E|$

$\therefore P(\text{full house}) = \frac{13\binom{4}{3}12\binom{4}{2}}{\binom{52}{5}}$

CHOOSING WITH REPETITION

Recall: $\binom{n}{r}$ = # subsets of size r of a set of size n

= # ways of choosing r items out of n, without order + without repetition

What if repetition is allowed? How many ways can we choose r items out of n without order?

Let x_i = # times the ith element is chosen

Then, we have the condition: $\underbrace{x_i \geq 0,\ x_i \in \mathbb{Z}}_{①}$, $\underbrace{\sum_{i=1}^{n} x_i = r}_{②}$

We want to count # $(x_1, \ldots, x_n)$ that satisfy ① and ②

EXAMPLE

S elements: $[5]$, $r = 3$

If we choose 1, 2, 4, then we denote $(1, 1, 0, 1, 0)$

If we choose 2, 3, 2, then we denote $(0, 2, 1, 0, 0)$

STARS AND BARS DRAWING

Represent user choices w/ stars and use bars to separate the different elements.

E.g. $(1, 0, 2, 0, 0)$

*		**		
1	2	3	4	5

We can simplify our notation: $*|\ \ |**|\ \ |$

Notice, every such diagram is a sequence of $n-1$ bars + r stars

Shun/翔海(@shun4midx)

From a different POV, it's equivalent to choosing $n+r-1$ symbols with $n-1$ bars + r stars

$\therefore$ # ways of choosing r elements from n w/ repetition w/o order = # ways of choosing r stars out of $n+r-1$ stars and bars $= \binom{n+r-1}{r} = \binom{n+r-1}{n-1}$

EXAMPLE

There is a probability course with 77 students, but the professor chooses 60 to pass.

choices $= \binom{77}{60}$

But now, the professor needs to assign grades to the students, so he decides to distribute 4500 pts among the 77 students. # choices = ?

Let $X_i :=$ # pts for the ith student, we need $\sum_{i=1}^{77} X_i = 4500$, $X_i \geq 0 \Rightarrow n=77, r=4500$

$\therefore$ # solutions $= \binom{77+4500-1}{77-1}$

What if each student needs ≥ 10 points?

☆ rmb we can just subtract values for lower bounds

That's the same as saying $X_i \geq 10 \ \forall i$, so define $Y_i := X_i - 10$, $Y_i \geq 0 \ \forall i$

$\therefore \sum_{i=1}^{77} Y_i = (\sum_{i=1}^{77} X_i) - 770 = 3730$

$\therefore$ # sols $= \binom{77+3730-1}{77-1}$

AXIOMS OF PROBABILITY

Suppose we roll a fair die 100 times and are interested in the sum of the rolls.

What is the sample space?

$S = \{100, \ldots, 600\}$ (but not uniform)

$S = [6]^{100}$ (but too large)

RECALL

Sample Space = Set S of all possible outcomes (sometimes denoted by Ω)

depends on your choice

Event = A subset $E \subseteq S$ that we are interested in